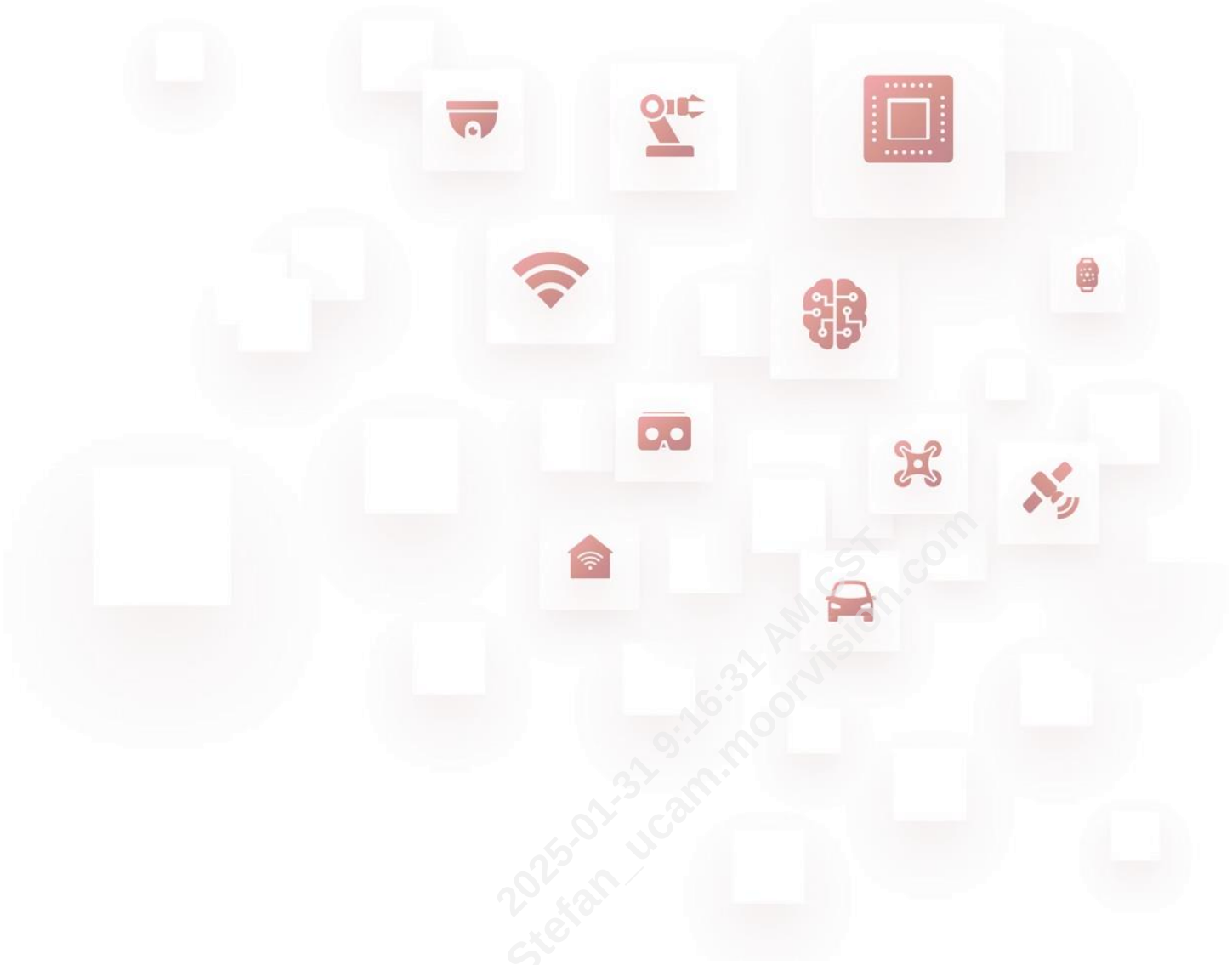
Thundercomm

Empowering Every IoT Device with Our Technology



2025-01-31 9:16:31 AM CST
Stefan_ucam.moorvision.com

Thundercomm TurboX™ C610 Open Kit

Display Panel Bringup Guide

Rev. V1.1
Jan 08, 2021

DN: TC_C610_6111

Revision History

Revision	Date	Description
1.0	Aug 07, 2020	Initial version
1.1	Jan 08, 2021	Update guidebook structure

Thundercomm Confidential
2025-01-31 9:16:31 AM CST
Stefan_ucam.moorvision.com

Copyright© 2021 Thundercomm Technology Co., Ltd. All rights reserved.

Table List

Table 2-1: Panel specification

Table 2-2: QCS610 reference board files

Table 2-3: QCS610 GPIO configuration reference files

Table 2-4: GPIO pins

Table 2-5: TurboX C610 power supplies reference files

Table 2-6: Reference power supplies

Thundercomm Confidential
2025-01-31 9:16:31 AM CST
Stefan_ucam.moorvision.com

Copyright© 2021 Thundercomm Technology Co., Ltd. All rights reserved.

Contents

Chapter 1. Introduction ..... - 2 -
1. Purpose ..... - 2 -
2. Conventions ..... - 2 -

Chapter 2. Panel Bringup in Kernel ..... - 3 -
1. Panel bringup workflow ..... - 3 -
2. Disable display panel in UEFI ..... - 5 -
3. Prepare panel.dtsi files ..... - 5 -
4. Calculate DSI PHY timing ..... - 5 -
5. Select and configure the panel ..... - 5 -
6. Configure GPIO pins ..... - 7 -
7. Panel power supplies ..... - 8 -
8. Configure the panel power ..... - 9 -

Chapter 3. Debug ..... - 12 -
1. Check display panel bind ..... - 12 -
2. Test display panel via modetest ..... - 12 -
3. Panel bringup in UEFI ..... - 13 -

Appendix 1. Intellectual Property Statement ..... - 14 -

Appendix 2. Trademarks ..... - 15 -

Thundercomm Confidential
2025-01-31 9:16:51 AM CST
Stefan_ucam.mobilevision.com

Chapter 1. Introduction

1. Purpose

This document provides guidelines for driver development and information on how to bring up the display module driver under LE (Linux Embedded) system on the Thundercomm TurboX™ C610 platform. The document also provides examples of the LCD panel configuration.

2. Conventions

- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, **cp armcc armcpp**.
- Code variables appear in angle brackets, for example, **<number>**.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font, for example:

  $ adb devices
- On the host computer use $ as shell prompt, for example:

  $ logcat
- On the target device use # as shell prompt, for example:

  # adb shell

Thundercomm Confidential
2025-01-31 9:16:31 AM CST
Stefan_ucam.moorvision.com

Chapter 2. Panel Bringup in Kernel

This chapter describes the kernel display panel bring up process on the QCS610 device and illustrates the process with a workflow.

Prerequisites:

Collect the following information from the panel specification and hardware schematic:

- Panel parameters – Panel type, resolution, bpp, fps, porch values
- Panel power-on sequence and signal duration for GPIO pins, for example, RESET/IOVDD
- Panel DSI initial command sequence and subsequent delay
- Bitclk to reach the target fps
- GPIO connection between the host and panel, including RESET, TE, and so on
- Panel power supplies and voltages

1. Panel bringup workflow

The following figure shows the workflow of the panel bringup in kernel based on the reference design.

Thundercomm Confidential
2025-01-31 9:16:31 AM CST
Stefan_ucam.moorvision.com

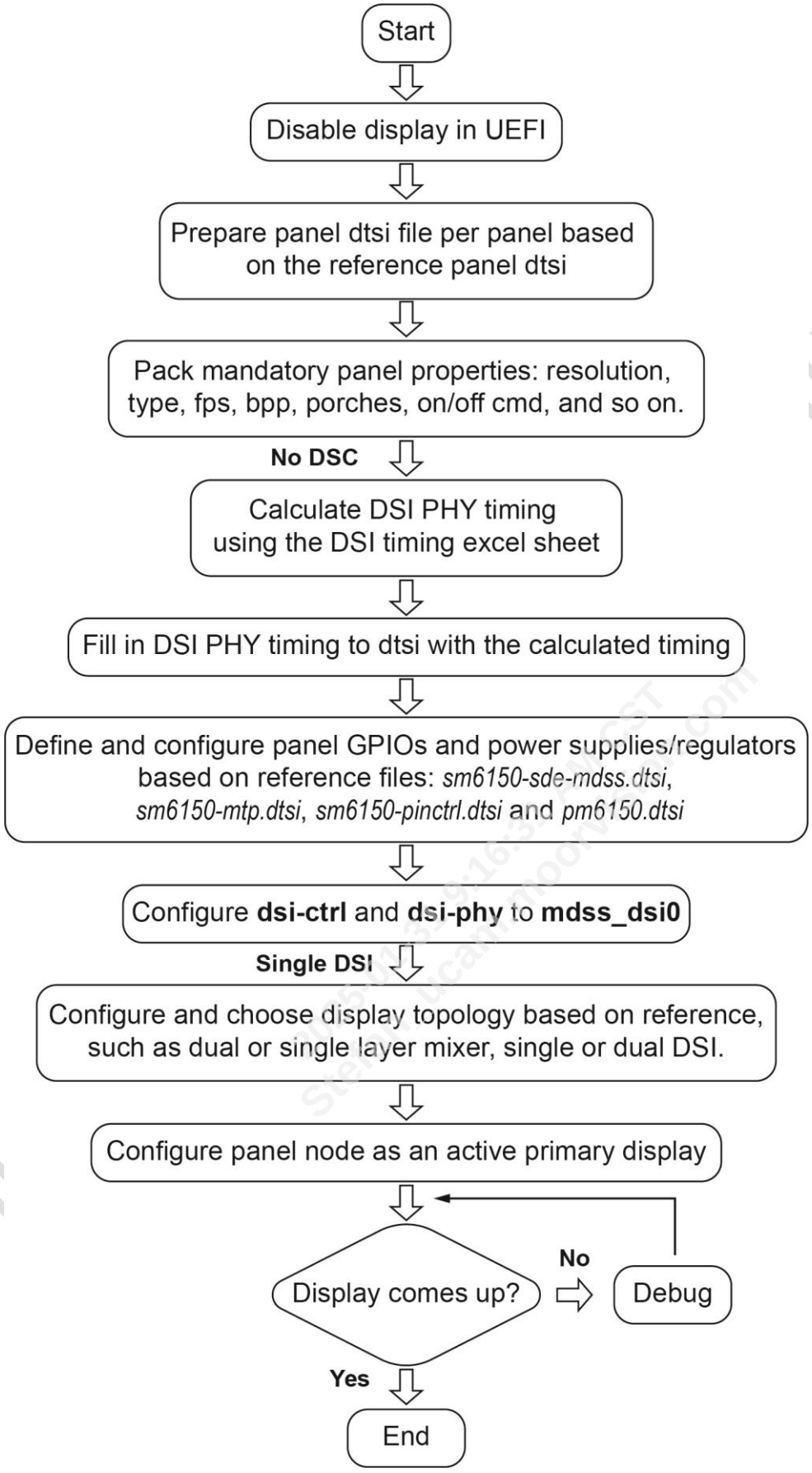


Figure 2-1. Kernel Panel Bringup Workflow

Following sections describe each of the steps in bringup process.

2. Disable display panel in UEFI

The debug support in UEFI (boot loader) is limited as compared to kernel. It is recommended to perform the panel bringup in the Linux kernel before the panel bringup in the UEFI. Disable the display panel in UEFI when doing the panel bringup in kernel.

The command to disable display panel in UEFI is:

```
#fastboot oem select-display-panel none
```

After the panel bringup in kernel, the same configurations can be used in UEFI. Usually, the panel name is passed to the kernel via a command line from UEFI. During the bringup in the kernel, as the display panel is disabled in UEFI, use pref-prim-pan dtsi entry to specify the panel name.

3. Prepare panel.dtsi files

The .dtsi files are based on QTI reference panels. The table lists all the supported panels on the internal device and the dtsi files are at *kernel/msm-4.14/arch/arm64/boot/dts/qcom/*.

Table 2-1: Panel specification

.dtsi file	Resolution	Panel type	DSI port	DSC
dsi-panel-ili988c-dual-video.dtsi	720x1280	Video mode	Single DSI	No

➲ **NOTE**: For a description of the device tree properties and mandatory elements, see kernel/msm-4.14/Documentation/devicetree/bindings/drm/msm/mdss-dsi-panel.txt.

4. Calculate DSI PHY timing

Use the *DSI Timing Parameters User Interactive Spreadsheet* (80-NH713-1) to calculate the PHY timing. The following is an example illustrating the calculation and mapping the PHY timing to the .dtsi file for ILI9881c Video mode (dsi-panel-ili988c-dual-video.dtsi) as the default IOT panel.

If you do not have a Qualcomm ID to access *DSI Timing Parameters User Interactive Spreadsheet* (80-NH713-1), email us at service@thundercomm.com to request PHY timing calculation and DTSI file configuration by providing panel porch parameters in email.

5. Select and configure the panel

The reference board files listed in the table are at: *kernel/msm-4.14/arch/arm64/boot/dts/qcom/*.

For information on DSI properties in the board file, refer to: */kernel/msm-4.14/Documentation/devicetree/bindings/drm/msm/mdss-dsi.txt*

Table 2-2: QCS610 reference board files

Thundercomm Confidential

Device name	Filename	Default panel type
IOT	qcs610-iot.dtsi	Video mode Default panel node: dsi_panel_ili988c_720p_vid_display

Follow the steps to configure the DSI properties and the panel information:

Step 1. Include the panel .dtsi file in the sm6150-sde-display.dtsi file at *kernel/msm-4.14/arch/arm64/boot/dts/qcom/*

This file includes all the supported panel .dtsi files.

```
#include "dsi-panel-yss99216oi050hd-720p-video.dtsi"
#include "dsi-panel-ili988c-dual-video.dtsi"
#include <dt-bindings/clock/mdss-14nm-pll-clk.h>
```

Step 2. Add a panel as DSI primary panel and define power and GPIO pins in dsi-panel-ili988c-dual-video.dtsi. Follow the example in the reference panel as follows:

1. Define display type as primary using “qcom,display-type”
2. Point to display panel node by “qcom,dsi-panel”
3. Define reset, TE, and other GPIO pins
4. Define power supplies, as per the following snippet for DSI interface Video mode panel:

```
dsi_ili9881c_720p_video: qcom,mdss_dsi_ili9881c_720p_video {
……
                qcom,mdss-dsi-panel-type = "dsi_video_mode";
                qcom,mdss-dsi-bl-pmic-control-type = "bl_ctrl_wled";
                qcom,mdss-dsi-bl-min-level = <1>;
                qcom,mdss-dsi-bl-max-level = <4095>;
                qcom,platform-reset-gpio = <&tlmm 91 0>;
                qcom,platform-te-gpio = <&tlmm 90 0>;
…..
};
```

Step 3. Add DSI PHY timing parameters and define display topology in the sm6150-sde-display.dtsi.

Define display topology based on panel information. See *Documentation/devicetree/bindings/drm/msm/mdss-dsi-panel.txt*

Display-topology has below parameters:

- Number of layer mixers
- Number of DSC compression encoders
- Number of DSI interfaces

Following is an example of FHD, where a non-DSC panel has supported topologies:

- Single layer mixer, none DSC, 1 DSI interfaces

```
&dsi_ili9881c_720p_video {
    qcom,mdss-dsi-t-clk-post = <0x0a>;
    qcom,mdss-dsi-t-clk-pre = <0x21>;
    qcom,mdss-dsi-display-timings {
        timing@0{
            qcom,mdss-dsi-panel-phy-timings = [
                1F 1C 04 06 03 02 04 0a
```

```
                                1F 1C 04 06 03 02 04 0a
                                1F 1C 04 06 03 02 04 0a
                                1F 1C 04 06 03 02 04 0a
                                1F 10 04 06 02 02 04 0a
                        ];
                        qcom,display-topology = <1 0 1>;
                        qcom,default-topology-index = <0>;
                };
        };
};
```

Step 4. Connect the DSI panel to the connector in sm6150-sde-display.dtsi by adding the panel name to “connectors” as shown in the code snippet. If needed, see *drivers/gpu/drm/msm/NOTES*.

```
&mdss_mdp {
        connectors = <&sde_wb &sde_dsi>;
};
```

6. Configure GPIO pins

Following are the steps to define and configure the GPIO pins:

Step 1. View the .dtsi files containing the GPIO configuration files used on QTI’s internal reference board. The files listed in the table are at *kernel/msm-4.14/arch/arm64/boot/dts/qcom/*.

Table 2-3: QCS610 GPIO configuration reference files

Panel Type	File name
IOT	sm6150-pinctrl.dtsi
Pinctrl	qcs610-pinctrl.dtsi dsi-panel-ili988c-dual-video.dtsi

The table lists the GPIO pins contained in the .dtsi files. The panel reset GPIO and TE GPIO I/O are defined in the panel node of the board .dtsi file or in dsi-panel-ili988c-dual-video.dtsi, and configured in sm6150-pinctrl.dtsi.

Table 2-4: GPIO pins

GPIO	Type	GPIO #
Panel reset	TLMM/MSM	GPIO 91
TE (Command mode panels only)	TLMM/MSM	GPIO 90

Step 2. Define and configure the reset and TE pins as shown in the following code *@kernel/msm-4.14/arch/arm64/boot/dts/qcom/dsi-panel-ili988c-dual-video.dtsi*.

```
qcom,platform-te-gpio = <&tlmm 90 0>; // TE pin
qcom,platform-reset-gpio = <&tlmm 91 0>; //Panel reset pin
```

Configure the reset pin in *@/arch/arm64/boot/dts/qcom/sm6150-pinctrl.dtsi*.

```
        pmx_sde_te {
                sde_te_active: sde_te_active {
```

```
                        mux {
                                pins = "gpio90";
                                function = "mdp_vsync";
                        };
                        config {
                                pins = "gpio90";
                                drive-strength = <2>;     /* 2 mA */
                                bias-pull-down;           /* PULL DOWN */
                        };
                };
                sde_te_suspend: sde_te_suspend {
                        mux {
                                pins = "gpio90";
                                function = "mdp_vsync";
                        };
                        config {
                                pins = "gpio90";
                                drive-strength = <2>;     /* 2 mA */
                                bias-pull-down;           /* PULL DOWN */
                        };
                };
        };
pmx_sde: pmx_sde {
                sde_dsi_active: sde_dsi_active {
                        mux {
                                pins = "gpio91";
                                function = "gpio";
                        };
                        config {
                                pins = "gpio91";
                                drive-strength = <8>;     /* 8 mA */
                                bias-disable = <0>;       /* no pull */
                        };
                };
                sde_dsi_suspend: sde_dsi_suspend {
                        mux {
                                pins = "gpio91";
                                function = "gpio";
                        };
                        config {
                                pins = "gpio91";
                                drive-strength = <2>;     /* 2 mA */
                                bias-pull-down;           /* PULL DOWN */
                        };
                };
        };
```

7. Panel power supplies

The regulators supplying power to the panel are defined in the panel .dtsi files. This enables support

of multiple DSI panels, which may require different sets of regulators on a single product. Because the regulators are a property of the board type, the qcom,supply_name binding maps the panel supply to the board regulator in qcs610-sde-display.dtsi.

The following tables list the reference power supplies in the TurboxC610 reference IOT projects. The .dtsi files are at *kernel/msm-4.14/arch/arm64/boot/dts/qcom/*.

Table 2-5: TurboX C610 power supplies reference files

Chip MDSS	File name	Description
qcs610 sde	sm6150-sde.dtsi	mdss_dsi0 – maps panel power supply to regulator mdss_dsi_phy0 – maps panel power supply to regulator
qcs610 panel .dtsi file	sm6150-sde-display.dtsi qcs610-iot.dtsi	Defines panel power supply
qcs610 regulator	sm6150-regulator.dtsi	Defines regulator initialization voltage

Table 2-6: Reference power supplies

Description	Name	Regulator	Voltage
Panel power supply 0	Vddio	PM6150/L11	1.8
Panel power supply 1	Lab	PM6150_L/lcdb_ldo_vreg	5.5
Panel power supply 2	Ibb	PM6150_L/lcdb_ncp_vreg	5.5
Panel power supply 3	Vdd3p	PM6150L_L7	3.0
DSI CTRL power	Vdda	PM6150/L1	1.2
DSI PHY power	Vcca	PM6150_L /L1	0.88

8. Configure the panel power

Define and configure the panel power supplies as follows:

Step 1. Specify qcom,panel-supply-entries in the panel .dtsi files as shown below in *kernel/arch/arm64/boot/dts/qcom/sm6150-sde-display.dtsi*.

There may be one or more supplies depending on panel requirements.

kernel/arch/arm64/boot/dts/qcom/dsi-panel-ili988c-dual-video.dtsi.

```
dsi_ili9881c_720p_video: qcom,mdss_dsi_ili9881c_720p_video {
.....
                qcom,panel-supply-entries = <&dsi_panel_pwr_supply>;
                qcom,mdss-dsi-bl-pmic-control-type = "bl_ctrl_wled";
.....
};
```

kernel/arch/arm64/boot/dts/qcom/sm6150-sde-display.dtsi

```
&soc {
    dsi_panel_pwr_supply: dsi_panel_pwr_supply {
            #address-cells = <1>;
            #size-cells = <0>;
```

Thundercomm Confidential

```
                qcom,panel-supply-entry@0 {
                        reg = <0>;
                        qcom,supply-name = "vddio";
                        qcom,supply-min-voltage = <1800000>;
                        qcom,supply-max-voltage = <1800000>;
                        qcom,supply-enable-load = <62000>;
                        qcom,supply-disable-load = <80>;
                        qcom,supply-post-on-sleep = <20>;
                };

                qcom,panel-supply-entry@1 {
                        reg = <1>;
                        qcom,supply-name = "lab";
                        qcom,supply-min-voltage = <4600000>;
                        qcom,supply-max-voltage = <6000000>;
                        qcom,supply-enable-load = <100000>;
                        qcom,supply-disable-load = <100>;
                };

                qcom,panel-supply-entry@2 {
                        reg = <2>;
                        qcom,supply-name = "ibb";
                        qcom,supply-min-voltage = <4600000>;
                        qcom,supply-max-voltage = <6000000>;
                        qcom,supply-enable-load = <100000>;
                        qcom,supply-disable-load = <100>;
                        qcom,supply-post-on-sleep = <20>;
                };

                qcom,panel-supply-entry@3 {
                        reg = <3>;
                        qcom,supply-name = "vdd3p";
                        qcom,supply-min-voltage = <3000000>;
                        qcom,supply-max-voltage = <3312000>;
                        qcom,supply-enable-load = <100000>;
                        qcom,supply-disable-load = <80>;
                        qcom,supply-pre-on-sleep = <20>;
                        qcom,supply-post-off-sleep = <20>;
                };
        };
.....
};
```

Step 2. Map the supply to the regulator in the board file by specifying the panel supply regulator in the corresponding display node as shown below in */arch/arm64/boot/dts/qcom/sm6150-sde-display.dtsi*.

```
sde_dsi: qcom,dsi-display {
……
                vddio-supply = <&pm6150_l13>;
                vdda-3p3-supply = <&pm6150_l18>;
```

```
                lab-supply = <&lcdb_ldo_vreg>;
                ibb-supply = <&lcdb_ncp_vreg>;
                vdd3p-supply = <&pm6150l_l7>;
……
}
```

➲ **NOTE**: While panel power supplies may be panel-specific, DSI CORE, DSI controller, and PHY are common resources shared across panels. The following example is for reference only at */arch/arm64/boot/dts/qcom/sm6150-sde.dtsi*.

```
mdss_dsi0: qcom,mdss_dsi_ctrl0@ae94000 {
……
                vdda-1p2-supply = <&pm6150l_l3>;
……
                qcom,ctrl-supply-entries {
                        #address-cells = <1>;
                        #size-cells = <0>;
                        qcom,ctrl-supply-entry@0 {
                                reg = <0>;
                                qcom,supply-name = "vdda-1p2";
                                qcom,supply-min-voltage = <1232000>;
                                qcom,supply-max-voltage = <1232000>;
                                qcom,supply-enable-load = <21800>;
                                qcom,supply-disable-load = <0>;
                        };
                };
……
};

mdss_dsi_phy0: qcom,mdss_dsi_phy0@ae94400 {
……
                vdda-0p9-supply = <&pm6150_l4>;
……
                qcom,phy-supply-entries {
                        #address-cells = <1>;
                        #size-cells = <0>;
                        qcom,phy-supply-entry@0 {
                                reg = <0>;
                                qcom,supply-name = "vdda-0p9";
                                qcom,supply-min-voltage = <880000>;
                                qcom,supply-max-voltage = <975000>;
                                qcom,supply-enable-load = <36000>;
                                qcom,supply-disable-load = <0>;
                        };
                };
……
};
```

Step 3. For more information on the DSI controller and its properties, see */Documentation/devicetree/bindings/drm/msm/mdss-dsi.txt*.

See Configure the panel power section if the panel uses different regulators in its hardware layout or needs more power supplies.

Chapter 3. Debug

This chapter describes the display panel debugging guideline.

1. Check display panel bind

You can check the LCD panel bind in the system via dmesg, following is bind successful log

```
$ adb shell
/ # dmesg   | grep dsi
[      3.649109] mdss_pll ae94400.qcom,mdss_dsi_pll: Failed to create debugfs directory
[      3.649135] mdss_pll ae94400.qcom,mdss_dsi_pll: Failed to create debugfs directory
[      3.649156] mdss_pll ae94400.qcom,mdss_dsi_pll: Failed to create debugfs directory
[      3.650543] dsi_pll_clock_register_14nm: Registered DSI PLL ndx=0 clocks successfully
[      7.055691] dsi_phy ae94400.qcom,mdss_dsi_phy0: ae94400.qcom,mdss_dsi_phy0 supply gdsc not found, using dummy regulator
[      7.067086] msm-dsi-phy:[dsi_phy_driver_probe] Probe successful for dsi-phy-0
[      7.075780] drm_dsi_ctrl ae94000.qcom,mdss_dsi_ctrl0: ae94000.qcom,mdss_dsi_ctrl0 supply refgen not found, using dummy regulator
[      7.103356] dsi-ctrl:[dsi_ctrl_dev_probe] Probe successful for dsi-ctrl-0
[      7.112159] msm-dsi-display:[dsi_display_parse_te_data] invalid vsync source selection
[      7.120315] OF: graph: no port node found in /soc/qcom,dsi-display
[      7.126686] msm-dsi-panel:[dsi_panel_parse_triggers:1129] [ILI988C video signal panel] fallback to default te-pin-select
[      7.137868] msm-dsi-panel:[dsi_panel_parse_panel_mode:1644] dsi_panel_parse_panel_mode: panel operating mode switch feature disabled
[      7.150109] msm-dsi-panel:[dsi_panel_parse_qsync_caps:1327] [ILI988C video signal panel] qsync min fps not defined rc:-22
[      7.161382] msm-dsi-panel:[dsi_panel_parse_reset_sequence:2002] RESET SEQ LENGTH = 24
[      7.169498] msm-dsi-panel:[dsi_panel_parse_misc_features:2052] dsi_panel_parse_misc_features: ulps feature disabled
[      7.180231] msm-dsi-panel:[dsi_panel_parse_misc_features:2058] dsi_panel_parse_misc_features: ulps during suspend feature disabled
[     11.002016] msm-dsi-display:[dsi_display_bind] Successfully bind display panel 'dsi_panel_ili988c_720p_vid_display'
[     11.012976] msm_drm ae00000.qcom,mdss_mdp: bound soc:qcom,dsi-display (ops dsi_display_comp_ops)
```

2. Test display panel via modetest

Step 1. Set up the **modetest** on the Yocto project.

a. Export machine distro

```
cd <project_root>/apps_proc/
MACHINE=qcs610-odk DISTRO=qti-distro-fullstack-debug VARIANT=debug
source poky/qti-conf/set_bb_env.sh
```

b. Modified local.conf

```
vim conf/local.conf
- INHERIT += "rm_work"
```

```
+ #INHERIT += "rm_work"
```

c. build libdrm

```
bitbake -fc cleanall libdrm ; bitbake libdrm
```

d. Set up the **modetest** on the device

```
cd    tmp-glibc/work/armv7ahf-neon-oe-linux-gnueabi/libdrm/2.4.83-r0/package/
tar -zcvf modetest.tar usr/
adb root
adb shell mount -o remount,rw /
adb push modetest.tar /
tar -xvf modetest.tar
```

Step 2. Test the display screen by **modetest**

Enter following command to test display panel

```
$ adb shell
/ # modetest -M msm_drm | more
Encoders:
id      crtc    type    possible crtcs  possible clones
27      0       DSI     0x00000007      0x00000000
43      0       Virtual 0x00000007      0x00000000
52      0       TMDS    0x00000007      0x00000000
57      0       DPMST   0x00000007      0x00000000
58      0       DPMST   0x00000007      0x00000000

Connectors:
id      encoder status          name            size (mm)       modes   encoders
28      0       connected       DSI-1           62x110          1       27
  modes:
        name refresh (Hz) hdisp hss hse htot vdisp vss vse vtot)
  720x1280x60x82404 60 720 920 928 1028 1280 1304 1312 1336 flags: ; type: preferred
  props:
        1 EDID:
                flags: immutable blob
                blobs:

                value:
                        00ffffffffffff00446d010001000000
                        1b10010380502d780a0dc9a057479827
/ # modetest -M msm_drm -s 28:720x1280x60x82404
setting mode 720x1280x60x82404-60Hz@XR24 on connectors 28, crtc 107
```

3. Panel bringup in UEFI

As the startup logo shall open in UEFI, bring up the display panel in UEFI before powering on your device.

Thundercomm cannot provide UEFI code due to Qualcomm ID problem. If you want to bring up display panel, contact us at: service@thundercomm.com.

Appendix 1. Intellectual Property Statement

This document itself and the proprietary technical information contained herein is the property of Thundercomm. Without prior written authority of Thundercomm, no one shall copy, modify, or reprint the whole or part of this document. Offenders are liable for all related damages, loss, and litigations.

Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation.

Thundercomm Confidential
2025-01-31 9:16:31 AM CST
Stefan_ucam.moorvision.com

Appendix 2. Trademarks

Hexagon is a trademark of Qualcomm Technologies, Inc. and/or its affiliated companies. Arm is a registered trademark of Arm Limited (or its subsidiaries) in the US and/or elsewhere. All other trademarks are the property of their respective owners.

Thundercomm Confidential
2025-01-31 9:16:31 AM CST
Stefan_ucam.moorvision.com